

COPK	228	sea 50	2	27.4	0.905	0.137	0.054	0.013
COPK	229	sea 50	2	28.5	2.07	0.232	0.100	0.028
COPK	230	sea 50	2	26.4	1.52	0.221	0.0953	0.027

S2. Summary of antennae (as NH3-N) excretion data

Taxa	Code	Species	Depth (m)	T (°C)	Excretion (μmol/h)	D (mm)	C (mg)	N (mg)	Reference
COPE	S1	<i>Calanus pomginius</i>	2	-1.0	0.107	1.043	0.41	0.12	Reids et al. (2001)
COPE	S2	<i>Calanoides acutus</i> C5	2	-0.2	0.019	0.366	0.19	0.01	
COPE	S3	<i>Calanoides acutus</i> C4, C5	2	-0.5	0.014	0.218	0.10	0.02	
COPE	S4	<i>Alcaluella gigas</i>	30	-1.7	0.113	1.080	0.50	0.11	
COPE	S5	<i>Mesocyclops edax</i>	2	-1.4	0.031	0.262	0.11	0.027	
COPE	S6	<i>Calanus finmarchicus</i> C5F	2	0.1	0.013	0.387	0.21	0.030	
COPE	S7	<i>Calanus finmarchicus</i> C5F	30	2.3	0.030	0.454	0.22	0.020	
COPE	S8	<i>Calanus hyperboreus</i> C5F	30	1.3	0.049	3.95	2.34	0.26	
COPE	S9	<i>Calanus hyperboreus</i> C5 (Biom)	30	0.9	0.051	1.94	1.12	0.14	
COPE	S10	<i>Calanus hyperboreus</i> C5 (Biom)	30	0.3	0.030	2.077	1.43	0.18	
COPE	S11	<i>Mesocyclops edax</i> C5F	30	0.1	0.023	0.553	0.17	0.037	
COPE	S12	<i>Neocalanus plumbeus</i> C4	2	7.3	0.017	0.765	0.45	0.063	
COPE	S13	<i>Neocalanus plumbeus</i> C4	2	5.6	0.007	0.263	0.15	0.021	
COPE	S14	<i>Neocalanus plumbeus</i>	2	15.1	0.066	0.294	0.14	0.026	
COPE	S15	<i>Neocalanus cristatus</i>	2	6.3	0.164	1.39	0.79	0.13	
COPE	S16	<i>Eucalanus huxleyi</i> C5F	2	6.0	0.058	1.01	0.40	0.10	
COPE	S17	<i>Paracalanus crassirostris</i>	2	8.8	0.002	0.010	0.003	0.004	
COPE	S18	<i>Mesocyclops edax</i>	2	8.2	0.013	0.166	0.0654	0.0164	
COPE	S19	<i>Mesocyclops edax</i>	2	13.0	0.023	0.094	0.0433	0.013	
COPE	S20	<i>Acartia clausi</i>	2	14.8	0.086	0.0053	0.003	0.000	
COPE	S21	<i>Acartia longiremis</i>	2	9.6	0.002	0.001	0.0005	0.000	
COPE	S22	<i>Teretia discoidalis</i>	2	7.0	0.008	0.057	0.025	0.006	
COPE	S23	<i>Centropages poeppigi</i> C5F	2	22.0	0.014	0.054	0.04	0.003	
COPE	S24	<i>Neocalanus gracilis</i>	2	19.7	0.102	0.500	0.234	0.051	
COPE	S25	<i>Undulula vulgaris</i>	2	23.5	0.074	0.165	0.077	0.018	
COPE	S26	<i>Acartia clausi</i> C5F	2	22.0	0.011	0.021	0.009	0.002	
COPE	S27	<i>Paracalanus parvus</i>	2	22.0	0.015	0.034	0.014	0.003	
COPE	S28	<i>Eucalanus subsecutus</i> C5F	2	22.0	0.024	0.105	0.043	0.009	
COPE	S29	<i>Eucalanus acutus</i>	2	24.0	0.061	0.291	0.121	0.028	
COPE	S30	<i>Centropages furcatus</i> C5M	2	22.0	0.011	0.017	0.014	0.003	
COPE	S31	<i>Centropages brevis</i>	2	17.3	0.011	0.020	0.0082	0.0020	
COPE	S32	<i>Temora turbinata</i>	2	22.0	0.005	0.016	0.0066	0.0015	
COPE	S33	<i>Calanoides ellipticus</i> C5F	2	22.0	0.017	0.017	0.002	0.007	
COPE	S34	<i>Labidocera sp.</i>	2	22.0	0.070	0.184	0.076	0.018	
COPE	S35	<i>Teretia discoidalis</i>	2	22.0	0.010	0.026	0.012	0.003	
COPE	S36	<i>Neocalanus acutus</i> C5F	2	22.0	0.006	0.017	0.0049	0.0011	
COPE	S37	<i>Acartia tonsa</i>	2	22.0	0.008	0.072	0.030	0.007	
COPE	S38	<i>Cyclops bicus</i>	2	22.0	0.013	0.052	0.013	0.003	
COPE	S39	<i>Neocalanus minor</i>	2	26.9	0.021	0.040	0.017	0.005	
COPE	S40	<i>Undulula vulgaris</i>	2	27.6	0.118	0.067	0.030	0.005	
COPE	S41	<i>Eucalanus subsecutus</i> C5F	2	27.7	0.027	0.069	0.029	0.007	
COPE	S42	<i>Eucalanus subsecutus</i> C5F	2	24.0	0.023	0.104	0.043	0.010	
COPE	S43	<i>Eucalanus subsecutus</i> C5F	2	27.4	0.070	0.137	0.054	0.013	
COPE	S44	<i>Paracalanus acutus</i>	2	26.0	0.004	0.017	0.007	0.002	
COPE	S45	<i>Acartia clausi</i> C5F	2	28.7	0.022	0.010	0.004	0.001	
COPE	S46	<i>Temora turbinata</i>	2	27.0	0.000	0.002	0.000	0.000	
COPE	S47	<i>Centropages furcatus</i> C5M	2	27.8	0.016	0.019	0.008	0.002	
COPE	S48	<i>Calanoides ellipticus</i> C5F	2	26.0	0.020	0.066	0.023	0.005	
COPE	S49	<i>Calanoides ellipticus</i> C5F	2	27.4	0.024	0.049	0.020	0.005	
COPE	S50	<i>Labidocera acuta</i>	2	24.0	0.164	0.233	0.097	0.022	
COPE	S51	<i>Labidocera acuta</i>	2	28.5	0.147	0.228	0.098	0.028	
COPE	S52	<i>Labidocera acuta</i>	2	26.4	0.139	0.221	0.095	0.027	
COPE	S53	<i>Labidocera sp.</i> C5F	2	28.5	0.057	0.024	0.010	0.002	
COPE	S54	<i>Acartia tonsa</i>	2	25.0	0.004	0.001	0.004	0.001	
COPE	S55	<i>Acartia tonsa</i>	2	27.0	0.004	0.017	0.007	0.002	
COPE	S56	<i>Acartia tonsa</i> C5F	2	27.8	0.012	0.008	0.008	0.002	
COPE	S57	<i>Teretia discoidalis</i>	2	26.0	0.010	0.024	0.010	0.002	
COPE	S58	<i>Teretia discoidalis</i>	2	26.0	0.003	0.013	0.0053	0.0012	
COPE	S59	<i>Teretia discoidalis</i>	25	25.0	0.180	0.100	0.0566	0.016	
COPE	S60	<i>Undulula vulgaris</i>	25	25.5	0.080	0.220	0.099	0.026	
COPE	S61	<i>Paracalanus acutus</i>	1	23.0	0.019	0.072	0.032	0.008	
COPE	S62	<i>Macrobrachium</i>	1	25.0	0.029	0.066	0.024	0.006	
COPE	S63	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S64	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S65	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S66	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S67	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S68	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S69	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S70	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S71	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S72	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S73	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S74	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S75	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S76	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S77	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S78	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S79	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S80	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S81	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S82	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S83	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S84	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S85	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S86	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S87	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S88	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S89	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S90	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S91	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S92	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S93	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S94	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S95	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S96	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S97	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S98	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S99	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S100	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S101	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S102	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S103	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S104	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S105	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S106	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S107	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S108	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S109	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S110	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S111	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S112	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S113	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S114	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S115	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S116	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S117	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S118	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S119	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S120	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S121	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S122	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S123	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S124	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S125	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S126	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S127	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S128	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S129	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S130	<i>Obolus nivalis</i>	1	24.0	0.002	0.000	0.000	0.000	
COPE	S131	<i>Obolus nivalis</i>	1						

S3. Summary of ON ratio data

Taxa	Code	Species	Depth (m)	T°C	ON (by atom)	DW(mg)	C (mg)	N (mg)	Reference
COPE	S1	<i>Calanus propinquus</i>	2	-1.0	13.8	1.04	0.409	0.118	Bea et al. (2001)
	S2	<i>Calanoides acutus</i> CS	2	-0.2	13.6	0.794	0.303	0.033	
	S3	<i>Calanoides acutus</i> C4 CS	2	-0.5	12.0	0.218	0.096	0.018	
	S4	<i>Rhincalanus gigas</i>	50	-1.7	4.8	1.08	0.303	0.106	
	S5	<i>Mesocyclops edax</i>	2	-1.4	12.9	0.265	0.106	0.027	
	S6	<i>Calanus finmarchicus</i> C6F	2	0.1	30.6	0.387	0.215	0.030	
	S7	<i>Calanus glacialis</i> C6F	50	2.3	25.7	0.474	0.223	0.050	
	S8	<i>Calanus hyperboreus</i> C6F	50	1.3	37.9	3.59	2.338	0.253	
	S9	<i>Calanus hyperboreus</i> CS (bloom)	50	0.9	25.4	1.94	1.117	0.144	
	S10	<i>Calanus hyperboreus</i> CS (late bloom)	50	-0.3	44.5	2.68	1.634	0.180	
	S11	<i>Metridia longa</i> C6F	50	0.1	22.1	0.353	0.168	0.037	
	S12	<i>Noecalanus plumchrus</i> C4	2	7.3	50.0	0.785	0.452	0.063	
	S13	<i>Noecalanus plumchrus</i> C4	2	5.6	48.8	0.263	0.148	0.021	
	S14	<i>Noecalanus plumchrus</i>	2	15.1	10.5	0.294	0.135	0.026	
	S15	<i>Noecalanus cristatus</i>	2	6.3	12.7	1.59	0.790	0.133	
	S16	<i>Eucalanus hungi</i> C6F	2	6.0	17.3	1.01	0.404	0.099	
	S17	<i>Paradicalanus elongatus</i>	2	8.6	21.9	0.016	0.053	0.004	
	S18	<i>Metridia pacifica</i>	2	8.2	35.6	0.148	0.065	0.016	
	S19	<i>Metridia pacifica</i>	2	11.0	14.7	0.094	0.043	0.011	
	S20	<i>Acartia clausi</i>	2	14.8	6.6	0.085	0.035	0.009	
	S21	<i>Acartia longiremis</i>	2	9.6	16.9	0.081	0.035	0.009	
	S22	<i>Tortanus discoidatus</i>	2	7.0	19.6	0.057	0.025	0.006	
	S23	<i>Noecalanus gracilis</i>	2	19.7	20.1	0.500	0.234	0.051	
	S24	<i>Udinus vulgaris</i>	2	23.5	16.3	0.185	0.077	0.018	
	S25	<i>Euchaeta marina</i>	2	24.0	27.4	0.291	0.121	0.028	
	S26	<i>Centropages brachius</i>	2	17.3	11.1	0.098	0.082	0.002	
	S27	<i>Acartia tonsa</i>	2	22.0	12.5	0.072	0.030	0.007	
	S28	<i>Nannocalanus minor</i>	2	26.9	12.7	0.040	0.017	0.005	
	S29	<i>Udinus vulgaris</i>	2	27.6	14.8	0.167	0.070	0.018	
	S30	<i>Eucalanus subsecus</i> C6F	2	24.0	39.5	0.104	0.043	0.010	
	S31	<i>Eucalanus atrovirens</i>	2	27.4	16.2	0.137	0.054	0.013	
	S32	<i>Calanopia elliptica</i> C6F	2	26.0	17.4	0.056	0.023	0.005	
	S33	<i>Labidocera acuta</i>	2	24.0	14.4	0.233	0.097	0.022	
	S34	<i>Labidocera acuta</i>	2	28.5	17.6	0.228	0.098	0.026	
	S35	<i>Labidocera neri</i>	2	26.4	13.6	0.221	0.095	0.027	
	S36	<i>Acartia australis</i>	2	25.0	15.5	0.091	0.038	0.009	
	S37	<i>Tortanus gracilis</i> C6M	2	26.0	17.8	0.054	0.028	0.002	
	S38	<i>Labidocera farrani</i>	25	25.0	13.5	0.160	0.057	0.016	Bea and McKinnon (2012)
	S39	<i>Udinus vulgaris</i>	25	25.5	17.8	0.220	0.094	0.026	
	S40	<i>Pseudocalanus melale</i>	2	21.0	20.7	0.072	0.032	0.008	
	S41	<i>Macrocystella gracilis</i>	1	25.0	18.2	0.065	0.024	0.005	
	S42	<i>Oithona nictidis</i>	1	24.0	11.8	0.012	0.004	0.001	
EUPH	1a	see S4	600	5.5	41.4	60.5	23.5	5.3	Bea (2013a)
	2a	see S4	10	-1.6	42.0	30.8	14.0	3.4	
	2b	see S4	5	-1.3	34.5	9.10	3.82	1.06	
	3	see S4	26	19.1	1.89	0.78	0.20	0.20	
	4	see S4	2	27	16.3	3.54	1.40	0.38	
	5	see S4	1	28	22.6	3.29	1.37	0.37	
	7	see S4	2	19.7	14.5	0.207	0.117	0.029	
	8	see S4	40	12.5	15.4	5.00	2.02	0.54	
	9	see S4	1	28	11.3	1.82	0.74	0.21	
	10a	see S4	2	13	23.1	4.97	2.06	0.54	
	10b	see S4	100	5	90.0	12.8	4.6	1.2	
	10c	see S4	100	5	76.0	11.5	4.0	1.1	
	11a	see S4	35	-1.1	16.6	86.3	38.4	8.8	
	11b	see S4	3	-1.7	14.1	16.4	4.8	4.8	
	11c	see S4	35	-1.1	17.3	247.1	115.9	24.7	
	11d	see S4	3	-1.7	29.3	80.5	32.5	9.6	
	11e	see S4	3	-1.7	31.9	97.7	39.5	11.4	
	12	see S4	2	27	13.9	0.77	0.31	0.08	
	13a	see S4	100	-0.8	12.4	23.0	9.5	2.7	
	13b	see S4	0	20.0	82.9	39.2	8.9		
	14a	see S4	250	4	42.0	74.1	25.9	7.6	
	14b	see S4	100	12	4.7	32.59	12.8	3.7	
	16	see S4	4	15.7	45.4	17.0	5.0	5.0	
	17	see S4	2	15	21.6	2.00	0.87	0.232	
	17	see S4	50	1	44.3	17.3	8.9	1.5	
	18a	see S4	2	-0.4	26.3	0.47	0.20	0.05	
	18b	see S4	2	-1.7	27.1	5.49	2.23	0.54	
	18c	see S4	300	-0.46	22.3	6.89	2.96	0.83	
	20	see S4	25	20	5.00	2.10	0.55		
	21a	see S4	100	5	42.0	12.2	4.5	1.2	
	21b	see S4	100	5	35.0	7.02	2.60	0.71	
	26	see S4	1	26	30.7	12.8	5.4	1.6	
AMPH	13	see S4	2	27	26.3	2.44	0.99	0.18	Bea (2013b)
	3	see S4	2	26	15.9	1.62	0.41	0.07	
	5	see S4	3	-0.8	14.4	105.3	49.2	7.5	
	7a	see S4	28	13.7	11.98	5.86	0.87		
	7b	see S4	2	27	10.8	17.67	4.26	0.97	
	8a	see S4	2	27	12.8	9.48	2.26	0.48	
	8b	see S4	20	17.4	7.29	1.74	0.37		
	9b	see S4	250	5	40.8	6.15	3.28	0.46	
	12	see S4	2	22	20.3	2.99	1.25	0.30	
	14	see S4	250	4	29.1	2.13	0.98	0.20	
	15a	see S4	2	-0.9	13.7	7.06	2.67	0.58	
	15c	see S4	2	15	16.2	1.00	0.38	0.08	
	16a	see S4	9	24.2	2.90	1.19	0.23		
	16b	see S4	250	5	14.1	2.65	1.23	0.23	
	17	see S4	50	-0.1	69.3	2.98	1.14	0.24	
	18	see S4	5	29.8	1.18	0.56	0.10		
	19	see S4	2	28	22.7	12.4	4.44	0.99	
	20	see S4	2	-1.1	18.4	12.14	4.90	0.99	
	22a	see S4	2	13	71.1	3.96	1.42	0.32	
	22b	see S4	250	5	66.9	5.90	2.17	0.40	
	28	see S4	55	-1.6	40.8	193.1	87.6	12.2	
	35	see S4	600	0.2	21.9	81.4	38.5	5.7	
DECA	21a	see S4	600	0.2	21.4	263	122.2	21.3	Bea (2013a)
	27b	see S4	600	0.2	15.5	339	156.1	27.1	
	31b	see S4	1	30	46.1	26.7	12.3	3.2	
	39	see S4	1	27.5	14.0	21.5	10.8	2.8	
	31a	see S4	1	24.5	24.7	6.49	2.99	0.77	Quetin et al. 1980
	S1	<i>Acanthephyra carinata</i>	600	5.5	14.9	840			
	S2	<i>Hymnodius frontalis</i>	600	5.5	12.2	400			
	S3	<i>Parapapilio sulcatifrons</i>	600	5.5	91.0	520			
	S4	<i>Papilio chacei</i>	100	5.5	17.2	340			
	S5	<i>Papilio emarginata</i>	600	5.5	21.4	790			
	S6	<i>Syndesmonia cristata</i>	600	5.5	8.1	740			
	S7	<i>Gonadus propinquus</i>	600	5.5	10.3	140			
	S8	<i>Sergestes setella</i>	600	5.5	23.0	720			
	S9	<i>Sergestes phoebus</i>	600	5.5	12.8	900			
MYSI	1	see S4	2	14.9	8.6	2.88	1.27	0.32	Bea (2013c)
	2	see S4	2	10	29.8	1.89	0.78	0.20	
	6	see S4	600	-1.6	18.7	171	75.3	17.9	
	12	see S4	600	5.5	19.5	280	143	18.3	
	15a	see S4	600	0.15	39.6	127	52.6	9.2	
	16	see S4	600	5.5	14.9	980	479	57.7	
	17	see S4	550	5.5	44.8	1040	512	66.0	
	19	see S4	1	14	21.1	0.82	0.32	0.10	
	20a	see S4	1	14	15.6	1.72	0.68	0.20	
	29b	see S4	1	16	20.5	4.20	1.70	0.47	
	29c	see S4	1	16	22.5	2.90	1.17	0.33	
	32a	see S4	1	28.5	8.4	0.16	0.069	0.017	
	32b	see S4	1	28.5	7.9	1.33	0.59	0.14	
	36	see S4	1	28.5	18.5	4.09	1.79	0.45	
	38a	see S4	2	27.8	11.7	1.47	0.59	0.15	
OSTR	S1	<i>Gigantocypris mulleri</i>	500	0.2	24.6	54.9	17.7	4.5	Bea (1988)
	S2	<i>Gigantocypris mulleri</i>	500	0.2	22.4	60.1	56.7	14.9	
	S3	<i>Gigantocypris mulleri</i>	500	-0.9	25.4	48.0	14.2	3.9	
	S4	<i>Gigantocypris mulleri</i>	500	-1	21.0	139.7	43.4	11.7	
CHAEL	S1	<i>Adiantum neglectum</i>	2	21	7.7	0.234	0.070	0.020	Bea and McKinnon (2012)
	S2	<i>Ferussacella hispida</i>	2	24	27.2	0.330	0.134	0.037	Bea (unpublished data)
	S3		2	26	6.8	0.100	0.041	0.011	Reise et al. (1970)
	S4	<i>Ferussacella robusta</i>	2	27	5.0	0.750	0.291	0.074	Bea (1974)
	S5	<i>Ferussacella ephra</i>	27	20.7	0.710	0.249	0.056	Bea (unpublished data)	
	S6	<i>Mesocyclops minus</i>	2	15	7.2	0.094	0.034	0.010	Bea (1974)
	S7	<i>Parasagitta elegans</i>	2	9	8.7	1.03	0.42	0.12	Bea (1974)
	S8		50	-0.4	8.1	4.50	1.73	0.47	Bea and Skjoldal (1989)
	S9	<i>Parasagitta gazellae</i>	100	-1	13.1	35.3	7.1	2.0	Bea and Kirkwood (1989)
	S10	<i>Sagitta bipunctata</i>	2	27.5	8.2	0.450	0		
S11	<i>Serranopsis serranoides</i>	2	27	8.1	0.730	0.263	0.072	Bea (1974)	
S12	<i>Zoeosagitta bialata</i> C minor	2	24	10.2	0.021	0.084	0.010	Bea and McKinnon (2012)	
CHNI	3a	see S4	2	8.5	24.8	3.9	1.02	0.37	Bea (2014a)
	3b	see S4	150	1.1	44.2	14.00	2.34	0.60	
	7a	see S4	9	17.8	20.9	16.4	5.4	5.4	
	7a	see S4	100	-0.7	17.6	136	12.9	3.3	
	15	see S4	2	27	18.7	2.27	0.16	0.07	

SI Data code and species: C = Copepodid, F = Female, M = Male, J = Juvenile, A = Adult, FG = Gravid female

Taxon	Data Code	Species	Stage/Sex
COPE	1	<i>Pachyphthalma pacifica</i>	CIF
	2	<i>Lucicutia bicolorata</i>	COM
	3	<i>Lucicutia grandis</i>	CSF
	4	<i>Meridia asymmetrica</i>	CSF
	5	<i>Meridia curticauda</i>	CSF
	6	<i>Meridia okhotensis</i>	CSF
	7	<i>Pleuronamma scutellata</i>	CSM
	8	<i>Pleuronamma xiphus</i>	CSF
	9	<i>Candacia columbae</i>	COM
	10	<i>Acetipops rostrata</i>	CSM
	11	<i>Chiridius pacificus</i>	CSF
	12	<i>Chiridius streptus</i>	CSF
	13	<i>Euchirella brevis</i>	CSF
	14	<i>Euchirella galeata</i>	CSF
	15	<i>Euchirella mesonensis</i>	CSF
	16	<i>Euchirella rostrata</i>	CSF
	17	<i>Euchirella runcata</i>	COM
	18	<i>Gadnus brevicaudus</i>	CSF
	19	<i>Gadnus robustus</i>	CSF
	20	<i>Gadnus tenuispinus</i>	CSF
	21	<i>Gadnus variabilis</i>	CSF
	22	<i>Pseudochirella pacifica</i>	CSM
	23	<i>Pseudochirella spinifera</i>	CSM
	24	<i>Undulacheta pacifica</i>	CSF
	25	<i>Paranacheta barbata</i>	CSF
	26	<i>Paranacheta bristovii</i>	CSF
	27	<i>Paranacheta elongata</i>	COM
	28	<i>Paranacheta rubra</i>	CSM
	29	<i>Paranacheta pseudomutabilis</i>	COM
	30	<i>Paranacheta rubra</i>	CSM
	31	<i>Amalothrix valida</i>	CSF
	32	<i>Mitocalanus robustus</i>	CSF
	33	<i>Scaphocalanus medius</i>	CSF
	34	<i>Scaphocalanus medius</i>	CSF
	35	<i>Pachyphthalma pacifica</i>	CSF
	36	<i>Lucicutia bicolorata</i>	COM
	37	<i>Lucicutia grandis</i>	CSF
	38	<i>Meridia asymmetrica</i>	CSF
	39	<i>Meridia curticauda</i>	CSF
	40	<i>Meridia okhotensis</i>	CSF
	41	<i>Bradydium pacificus</i>	CSF
	42	<i>Bradydium pacificus</i>	CSF
	43	<i>Gadnus brevicaudus</i>	CSF
	44	<i>Gadnus robustus</i>	CSF
	45	<i>Gadnus tenuispinus</i>	CSF
	46	<i>Gadnus variabilis</i>	CSF
	47	<i>Pseudochirella pacifica</i>	CSM
	48	<i>Pseudochirella spinifera</i>	CSM
	49	<i>Undulacheta major</i>	COM
	50	<i>Undulacheta major</i>	COM
	51	<i>Undulacheta phasma</i>	CSF
	52	<i>Paranacheta bristovii</i>	CSF
	53	<i>Paranacheta elongata</i>	COM
	54	<i>Paranacheta rubra</i>	CSM
	55	<i>Paranacheta pseudomutabilis</i>	COM
	56	<i>Paranacheta rubra</i>	CSM
	57	<i>Paranacheta pseudomutabilis</i>	COM
	58	<i>Paranacheta rubra</i>	CSM
	59	<i>Paranacheta pseudomutabilis</i>	COM
	60	<i>Paranacheta rubra</i>	CSM
	61	<i>Cornucalanus indicus</i>	CSF
	62	<i>Ochrocalanus magnus</i>	CSF
	63	<i>Amalothrix inornata</i>	CSM
	64	<i>Amalothrix parvivalda</i>	CSF
	65	<i>Lophothrix frontalis</i>	CSF
	66	<i>Scaphocalanus magnus</i>	CSF
	67	<i>Scaphocalanus medius</i>	CSF
	68	<i>Scaphocalanus medius</i>	CSF
	69	<i>Scaphocalanus medius</i>	CSF
	70	<i>Scaphocalanus medius</i>	CSF
	71	<i>Scaphocalanus medius</i>	CSF
	72	<i>Scaphocalanus medius</i>	CSF
	73	<i>Scaphocalanus medius</i>	CSF
	74	<i>Scaphocalanus medius</i>	CSF
	75	<i>Scaphocalanus medius</i>	CSF
	76	<i>Scaphocalanus medius</i>	CSF
	77	<i>Scaphocalanus medius</i>	CSF
	78	<i>Scaphocalanus medius</i>	CSF
	79	<i>Scaphocalanus medius</i>	CSF
	80	<i>Scaphocalanus medius</i>	CSF
	81	<i>Scaphocalanus medius</i>	CSF
	82	<i>Scaphocalanus medius</i>	CSF
	83	<i>Scaphocalanus medius</i>	CSF
	84	<i>Scaphocalanus medius</i>	CSF
	85	<i>Scaphocalanus medius</i>	CSF
	86	<i>Scaphocalanus medius</i>	CSF
	87	<i>Scaphocalanus medius</i>	CSF
	88	<i>Scaphocalanus medius</i>	CSF
	89	<i>Scaphocalanus medius</i>	CSF
	90	<i>Scaphocalanus medius</i>	CSF
	91	<i>Scaphocalanus medius</i>	CSF
	92	<i>Scaphocalanus medius</i>	CSF
	93	<i>Scaphocalanus medius</i>	CSF
	94	<i>Scaphocalanus medius</i>	CSF
	95	<i>Scaphocalanus medius</i>	CSF
	96	<i>Scaphocalanus medius</i>	CSF
	97	<i>Scaphocalanus medius</i>	CSF
	98	<i>Scaphocalanus medius</i>	CSF
	99	<i>Scaphocalanus medius</i>	CSF
	100	<i>Scaphocalanus medius</i>	CSF
	101	<i>Scaphocalanus medius</i>	CSF
	102	<i>Scaphocalanus medius</i>	CSF
	103	<i>Scaphocalanus medius</i>	CSF
	104	<i>Scaphocalanus medius</i>	CSF
	105	<i>Scaphocalanus medius</i>	CSF
	106	<i>Scaphocalanus medius</i>	CSF
	107	<i>Scaphocalanus medius</i>	CSF
	108	<i>Scaphocalanus medius</i>	CSF
	109	<i>Scaphocalanus medius</i>	CSF
	110	<i>Scaphocalanus medius</i>	CSF
	111	<i>Scaphocalanus medius</i>	CSF
	112	<i>Scaphocalanus medius</i>	CSF
	113	<i>Scaphocalanus medius</i>	CSF
	114	<i>Scaphocalanus medius</i>	CSF
	115	<i>Scaphocalanus medius</i>	CSF
	116	<i>Scaphocalanus medius</i>	CSF
	117	<i>Scaphocalanus medius</i>	CSF
	118	<i>Scaphocalanus medius</i>	CSF
	119	<i>Scaphocalanus medius</i>	CSF
	120	<i>Scaphocalanus medius</i>	CSF
	121	<i>Scaphocalanus medius</i>	CSF
	122	<i>Scaphocalanus medius</i>	CSF
	123	<i>Scaphocalanus medius</i>	CSF
	124	<i>Scaphocalanus medius</i>	CSF
	125	<i>Scaphocalanus medius</i>	CSF
	126	<i>Scaphocalanus medius</i>	CSF
	127	<i>Scaphocalanus medius</i>	CSF
	128	<i>Scaphocalanus medius</i>	CSF
	129	<i>Scaphocalanus medius</i>	CSF
	130	<i>Scaphocalanus medius</i>	CSF
	131	<i>Scaphocalanus medius</i>	CSF
	132	<i>Scaphocalanus medius</i>	CSF
	133	<i>Scaphocalanus medius</i>	CSF
	134	<i>Scaphocalanus medius</i>	CSF
	135	<i>Scaphocalanus medius</i>	CSF
	136	<i>Scaphocalanus medius</i>	CSF
	137	<i>Scaphocalanus medius</i>	CSF
	138	<i>Scaphocalanus medius</i>	CSF
	139	<i>Scaphocalanus medius</i>	CSF
	140	<i>Scaphocalanus medius</i>	CSF
	141	<i>Scaphocalanus medius</i>	CSF
	142	<i>Scaphocalanus medius</i>	CSF
	143	<i>Scaphocalanus medius</i>	CSF
	144	<i>Scaphocalanus medius</i>	CSF
	145	<i>Scaphocalanus medius</i>	CSF
	146	<i>Scaphocalanus medius</i>	CSF
	147	<i>Scaphocalanus medius</i>	CSF
	148	<i>Scaphocalanus medius</i>	CSF
	149	<i>Scaphocalanus medius</i>	CSF
	150	<i>Scaphocalanus medius</i>	CSF

Taxon	Data Code	Species	Stage/Sex	Tax
	151	<i>Scaphocalanus affinis</i>	CSF	
	152	<i>Scaphocalanus affinis</i>	CSF	
	153	<i>Scaphocalanus</i> sp.	CSF	
	154	<i>Scotocalanus scutellatus</i>	CSF	
	155	<i>Scotocalanus scutellatus</i>	CSF	
	156	<i>Bathycalanus bradyi</i>	CSF	
	157	<i>Spinocalanus magnus</i>	CSF	
	158	<i>Aetideopsis rostrata</i>	CSF	
	159	<i>Amalthea incerta</i>	CSM	
	160	<i>Amalthea incerta</i>	CSM	
	161	<i>Bathesca lamellata</i>	CSM	
	162	<i>Bathesca lamellata</i>	CSF	
	163	<i>Bathesca lamellata</i>	C	
	164	<i>Bathycalanus bradyi</i>	CSM	
	165	<i>Bohyscalanus bradyi</i>	C	
	166	<i>Benthocalanus pellitella</i>	CSF	
	167	<i>Chiridella abyssalis</i>	CSF	
	168	<i>Chiridella pycnostoma</i>	CSF	
	169	<i>Gastanus parvicornis</i>	CSF	
	170	<i>Gaidius pungens</i>	CSF	
	171	<i>Gastanus robustus</i>	CSF	
	172	<i>Meridia ornata</i>	CSM	
	173	<i>Meridia ornata</i>	CSF	
	174	<i>Meridia ornata</i>	CSF	
	175	<i>Mitocalanus robustus</i>	CSM	
	176	<i>Ochocalanus magnus</i>	CSF	
	177	<i>Ochocalanus magnus</i>	CSF	
	178	<i>Paracantha</i> sp.	C	
	179	<i>Pseudochiridella pycnostoma</i>	CSM	
	180	<i>Pseudochiridella polygona</i>	CSF	
	181	<i>Spinocalanus affinis</i>	CSF	
	182	<i>Spinocalanus magnus</i>	CSF	
	183	<i>Undeucheta incisa</i>	CSM	
	184	<i>Undeucheta incisa</i>	CSF	
	185	<i>Undeucheta plumosa</i>	CSF	
	186	<i>Undeucheta plumosa</i>	CSF	
	187	<i>Undeucheta plumosa</i>	CSF	
	188	<i>Xanthocalanus karlensis</i>	CSF	
	189	<i>Xanthocalanus karlensis</i>	CSF	
	190	<i>Calanus propinquus</i>		
	191	<i>Meridia gerlachei</i>		
	192	<i>Calanoides acutus</i>	C	
	193	<i>Calanoides acutus</i>	CS5	
	194	<i>Calanus foveolatus</i>	CSF	
	195	<i>Calanus glacialis</i>	CSF	
	196	<i>Calanus hyperboreus</i>	CSF	
	197	<i>Calanus hyperboreus</i>	CSF	
	198	<i>Calanus hyperboreus</i>	CSF	
	199	<i>Meridia longa</i>	CSF	
	200	<i>Rinculius eggs</i>		
	201	<i>Neocalanus crinitus</i>	CS	
	202	<i>Neocalanus plumosus</i>	C	
	203	<i>Ecalsanus hung</i>	CS	
	204	<i>Pseudocalanus elongatus</i>		
	205	<i>Meridia pacifica</i>	CSF	
	206	<i>Acartia longiremis</i>		
	207	<i>Tortania diluviana</i>		
	208	<i>Meridia pacifica</i>		
	209	<i>Mesocalanus tenuicornis</i>		
	210	<i>Paracalanus parvus</i>		
	211	<i>Centropages abdominalis</i>		
	212	<i>Pseudodiplopus marinus</i>		
	213	<i>Neocalanus plumosus</i>		
	214	<i>Acartia clausi</i>		
	215	<i>Neocalanus gracilis</i>		
	216	<i>Centropages brachioris</i>		
	217	<i>Acartia tonsa</i>		
	218	<i>Undulina vulgaris</i>		
	219	<i>Euchaeta maris</i>		
	220	<i>Acartia pacifica</i>		
	221	<i>Calanopia elliptica</i>		
	222	<i>Tortania gracilis</i>		
	223	<i>Ecalsanus subarcticus</i>		
	224	<i>Labidocera acuta</i>		
	225	<i>Acartia australis</i>		
	226	<i>Neomcalanus minor</i>		
	227	<i>Undulina vulgaris</i>		
	228	<i>Ecalsanus attenuatus</i>		
	229	<i>Labidocera acuta</i>		
	230	<i>Labidocera nerii</i>		
	231	<i>Pontella danae</i>		
	232	<i>Neocalanus crinitus</i>	CS	
	233	<i>Calanus marshallae</i>	CS	
	234	<i>Neocalanus plumosus</i>	CSF	
	235	<i>Neocalanus plumosus</i>	CSF	
	236	<i>Mesocalanus tenuicornis</i>	CSF	
	237	<i>Meridia pacifica</i>	CSF	
	238	<i>Meridia pacifica</i>	CSF	
	239	<i>Platommata scutellata</i>	CSF	
	240	<i>Platommata scutellata</i>	CSF	
	241	<i>Platommata scutellata</i>	CSF	
	242	<i>Karumanna tzipialis</i>	CSF	
	243	<i>Platommata tzipialis</i>	CSF	
	244	<i>Gastanus simplex</i>	CSF	
	245	<i>Gastanus simplex</i>	CSF	
	246	<i>Euchaeta maris</i>	CSF	
	247	<i>Euchaeta maris</i>	CSF	
	248	<i>Euchaeta maris</i>	CSF	
	249	<i>Euchaeta maris</i>	CSF	
	250	<i>Conductia bipinnata</i>	CSF	
251	<i>Conductia bipinnata</i>	CSF		
252	<i>Conductia bipinnata</i>	CSF		
253	<i>Conductia bipinnata</i>	CSF		
254	<i>Conductia bipinnata</i>	CSF		
EUPH	1	<i>Benthopellicularia antipodis</i>		
	1b	<i>Benthopellicularia amblyops</i>		
	1c	<i>Benthopellicularia amblyops</i>		
	2a	<i>Euphausia crystallorophias</i>		
	2b	<i>Euphausia crystallorophias</i>	J	
	3	<i>Euphausia distomedea</i>		
	4	<i>Euphausia distomedea</i>		
	5	<i>Euphausia gibbsii</i>		
	6	<i>Euphausia gibboides</i>		
	7	<i>Euphausia irrorata</i>		
	8	<i>Euphausia lucens</i>		
	9	<i>Euphausia rostris</i>		
	10a	<i>Euphausia pacifica</i>		
	10b	<i>Euphausia pacifica</i>	M	
	10c	<i>Euphausia pacifica</i>	F	
	11a	<i>Euphausia superba</i>		
	11b	<i>Euphausia superba</i>		
	11c	<i>Euphausia superba</i>	JG	
	11d	<i>Euphausia superba</i>	JG	
	11e	<i>Euphausia superba</i>	M	
12	<i>Euphausia tenera</i>			
13a	<i>Euphausia iracunda</i>			
13b	<i>Euphausia iracunda</i>			
14a	<i>Megacyclops bicus notus</i>			
14b	<i>Megacyclops bicus notus</i>			
14c	<i>Megacyclops bicus notus</i>			
15	<i>Niphanes antipodis</i>			
16	<i>Tessarakolpax oculatus</i>			
17	<i>Thysanoessa inermis</i>			
18a	<i>Thysanoessa inermis</i>	J		
18b	<i>Thysanoessa inermis</i>			
18c	<i>Thysanoessa inermis</i>			
19	<i>Thysanoessa inermis</i>			
20	<i>Thysanoessa sp.</i>			
21a	<i>Thysanoessa spinifera</i>	F		
21b	<i>Thysanoessa spinifera</i>	M		
22	<i>Thysanoessa setacea</i>			
23	<i>Thysanoessa monacantha</i>			
24	<i>Thysanoessa longicauda</i>			
DECA	1	<i>Acanthephyra aculeifrons</i>		
	2a	<i>Acanthephyra carinatastris</i>		
	2b	<i>Acanthephyra carinatastris</i>		
	2c	<i>Acanthephyra carinatastris</i>		
	3	<i>Acanthephyra cecilia</i>		
	4a	<i>Acanthephyra purpurata</i>		
	4b	<i>Acanthephyra purpurata</i>		
	4c	<i>Acanthephyra purpurata</i>		
	5	<i>Acanthephyra quadripinnata</i>		
	6	<i>Acanthephyra quadripinnata</i>		